MRP - Robust CNN-Based Binary Image Recognition System

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MSCS2201

Motivation, Main idea, and Results

- Motivation:
  - Explore how Convolutional Neural Networks (CNNs) can automatically classify images, using cat vs dog classification as a practical example. This task highlights key computer vision challenges like feature extraction and generalization, while providing a clear way to evaluate performance.
- Main Idea:
  - This project uses a Convolutional Neural Network (CNN) to automatically learn visual features and classify images of cats and dogs. A separate inference script loads the trained model to predict new images with confidence scores.
- Results:
  - The model trained for 12 epochs and achieved 98% accuracy on validation data, with high precision and recall for both classes.

Literature Review

- Deep learning dominance: CNNs have completely replaced traditional methods by automatically learning hierarchical features directly from raw pixels, eliminating manual feature engineering.
- Transfer learning standard: Pre-trained models (like MobileNetV2) consistently outperform scratch-trained networks by leveraging features learned from large datasets like ImageNet, drastically reducing training time and data requirements.
- MobileNetV2: is a lightweight CNN built for speed and efficiency on phones or low-power devices. It uses depthwise separable convolutions that splits the filtering process into two simpler steps, drastically reducing computation while keeping good accuracy.
- Trade-offs: Bigger models = more accurate but slower. Smaller models like MobileNetV2 = faster and use less battery, perfect for phones. Pick whatever is more efficient for the project.

Approach Used

- Dataset:
  - https://download.microsoft.com/download/3/E/1/3E1C3F21-ECDB-4869-8368-6DEBA77B919F/kagglecatsanddogs_5340.zip
- Data Preparation
  - Custom cleaning function using PIL to verify and remove corrupted images
  - image_dataset_from_directory with 80/20 validation split
- Model Architecture
  - Base: MobileNetV2 (pre-trained on ImageNet, frozen)
    - Contains ~53 convolutional layers (pre-built)
    - https://www.kaggle.com/models/google/mobilenet-v2/tensorFlow2/tf2-preview-classification/4?tfhub-redirect=true
  - Data augmentation: RandomFlip + RandomRotation

Approach Used

- Training
  - Transfer learning with frozen base model (non-trainable layers in MobileNetV2)
  - Loss: Binary crossentropy
  - Epochs: 12, Batch size: 32
- Total Parameters: MobileNetV2: ~2,250,000, Dense(64): 81,984, Dense(1): 65
  - Total ≈ 2.33 *million*
- Inference (test.py)
  - Loads trained model from .h5 file
  - Preprocesses single image: PIL load → RGB convert → resize (160×160)
  - Returns class (Cat/Dog) with confidence percentage

Code Used

- Pretrained CNN Backbone (MobileNetV2) ~53 convolutional layers used for feature extraction (ImageNet pretrained)

```
base_model = tf.keras.applications.MobileNetV2(
```

- CNN Layers Frozen (Transfer Learning) ~2.25M parameters are **non-trainable**, reducing computation and overfitting

```
base_model.trainable = False
```

- Custom Classification Layers (Trainable) - Only these layers learn → convert features into cat vs dog prediction

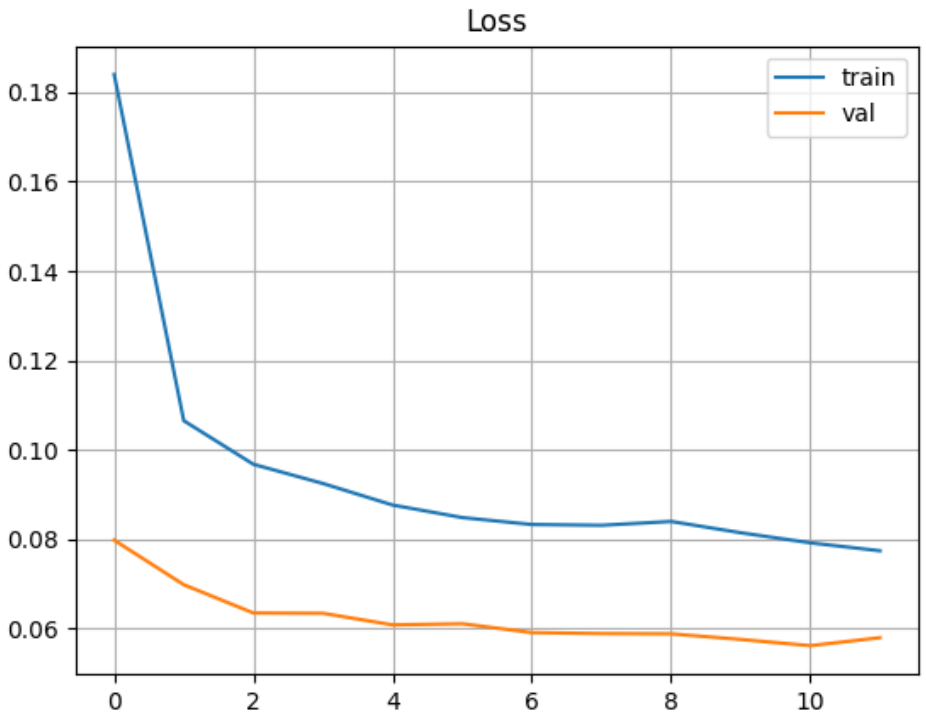
```
tf.keras.layers.GlobalAveragePooling2D(),
tf.keras.layers.Dense( units: 64, activation='relu'),
tf.keras.layers.Dense( units: 1, activation='sigmoid')
```

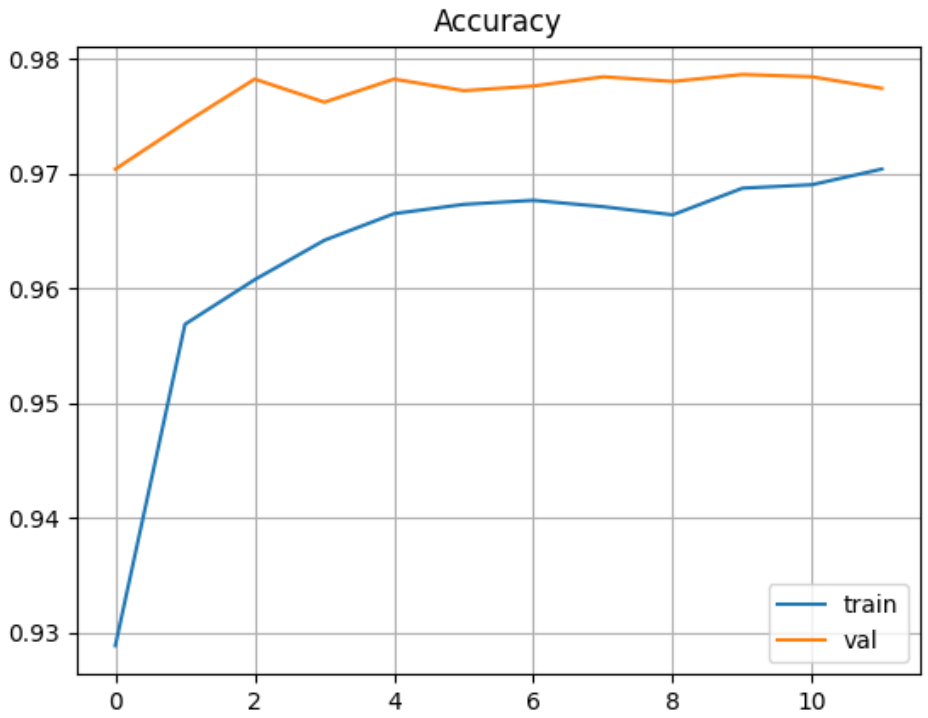
- **Total Parameters** ~2.3M total parameters - Majority from CNN (frozen) - Few thousand are trainable (Dense layers)
- Data Processing Layers - Normalize input and augment data to improve generalization

```
data_aug = tf.keras.Sequential([
    tf.keras.layers.RandomFlip("horizontal"),
    tf.keras.layers.RandomRotation(0.1),
])
```

- **Training Setup - Uses Adam optimizer, binary cross-entropy, trained for 12 epochs**

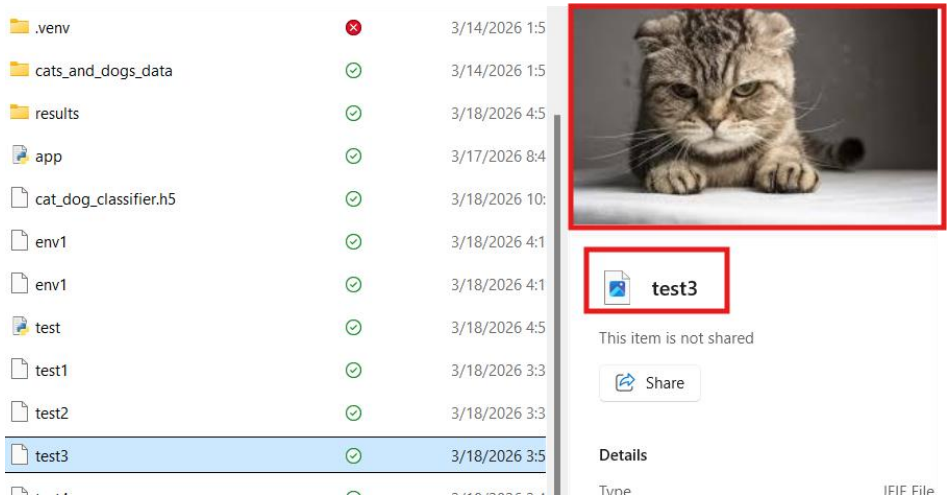
Results





- The pattern where training accuracy is slightly lower than validation accuracy indicates effective regularization through data augmentation. Since augmentation (RandomFlip and RandomRotation) is applied only to the training set, the model sees artificially varied versions of images during training, making learning intentionally harder—while validation evaluates on clean, original images.
- This prevents the model from simply memorizing pixel patterns and forces it to learn true distinguishing features of cats and dogs, resulting in better generalization. The 80/20 data split ensures both sets come from the same distribution, but the augmentation gap explains why validation performance can appear slightly better without indicating overfitting.

Results




```
Terminal    Local
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

(.venv) PS C:\Users\Marina\OneDrive\Desktop\Artificial Intelligence\MSP> python test.py test3.jfif
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1773884600.070912   25556 port.cc:153] oneDNN custom operations are on. You may see slightly d
ferent computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1773884632.899254   25556 port.cc:153] oneDNN custom operations are on. You may see slightly d
ferent computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even
TensorFlow-DirectML plugin.
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metr
test3.jfif -> Cat (99.99%)

(.venv) PS C:\Users\Marina\OneDrive\Desktop\Artificial Intelligence\MSP>
```

Links To Code and Video

- Github
  - https://github.com/marinarizkk/ai-hw-winter-2026-nn/tree/main/MRP
- YouTube
  - https://youtu.be/Gdfhgzh-f9o

THANK YOU!